

A NOTE ON A toy model of credit frictions

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1. The model

I develop here a toy model of credit market for firms that relies on the collateral constraint mechanism à la [Kiyotaki and Moore \(1997\)](#).

1.1 Households

Suppose a welfare maximizing household enjoying current and future consumption and dislike working with respective curvatures $\sigma_C, \sigma_H > 0$. The household maximizes the following stream of utilities discounted at a fixed rate β_H :

$$\max_{\{C_t^H, d_t, H_t\}} \sum_{\tau=0}^{\infty} \beta_H^\tau E_t \left\{ \frac{(C_{t+\tau}^H)^{1-\sigma_C}}{1-\sigma_C} - \chi \frac{H_{t+\tau}^{1+\sigma_H}}{1+\sigma_H} \right\} \quad (1)$$

where C_t^H is consumption, H_t is hours worked and $\chi > 0$ is a shift parameter allowing to pin down the amount of hours worked in steady state.

The budget constraint in real terms reads as:

$$C_t^H + d_t = R_{t-1}d_{t-1} + H_t w_t - T_t, \quad (2)$$

where d_t is the amount of real deposits, R_{t-1} is the real interest rate of real deposits d_t , w_t the real wage is computed as $w_t = W_t/P_t$ and a lump-sum tax financing the government T_t .

The Lagrangian problem reads as:

$$\begin{aligned} \mathcal{L}_t^H = & \sum_{\tau=0}^{\infty} \beta^\tau \left[\frac{(C_{t+\tau}^H)^{1-\sigma_C}}{1-\sigma_C} - \chi \frac{H_{t+\tau}^{1+\sigma_H}}{1+\sigma_H} \right] \\ & + \beta^\tau \lambda_{t+\tau}^H \left[\frac{R_{t-1+\tau}}{\pi_{t+\tau}} d_{t-1+\tau} + H_{t+\tau} w_{t+\tau} - C_{t+\tau}^H - d_{t+\tau} - T_{t+\tau} \right] \end{aligned}$$

where λ_t^H is a lagrangian multiplier.

First order conditions are:

$$C_t^H : (C_t^H)^{-\sigma_C} - \lambda_t^H = 0 \quad (3)$$

$$d_t : \beta_H E_t \{ \lambda_{t+1}^H \} R_t - \lambda_t^H = 0 \quad (4)$$

$$H_t : -\chi H_t^{\sigma_H} + \lambda_t^H w_t = 0 \quad (5)$$

1.2 Entrepreneurs

Let us now suppose that there is a large number of identical entrepreneurs which are credit constraint. They hire workers and combine them with capital (both produced by them and supplied by households). As in [Iacoviello \(2015\)](#), entrepreneurs are subject to a borrowing constraint that acts as a wedge on the capital and labor demand:

$$l_t = m_K E_t \{ q_{t+1} \} \frac{K_t}{r_t^E} - m_H h_t w_t \quad (6)$$

where l_t is the real stock of real loans, entrepreneurs cannot borrow more than a fraction $m_K \geq 0$ of the expected value of their real capital stock, given by $E_t \{ q_{t+1} \} K_t / r_t^E$. Here, q_{t+1} is the expected selling price of capital goods, if shocks between t and $t+1$ are too large, financial intermediaries must be strongly binding and generate a large recession. Variable r_t^E is the real borrowing rate. This was one of the key result of [Kiyotaki and Moore \(1997\)](#) (*i.e.* small shocks, large recessions). In addition, the borrowing constraint stipulates that a fraction $m_H \geq 0$ of the wage bill must be paid in advance.

In the same vein as for households, the welfare function discounted at a rate β_E ($< \beta_H$) reads as:

$$\max_{\{C_t^E, K_t, I_t, L_t, h_t\}} \sum_{\tau=0}^{\infty} \beta_E^\tau E_t \left\{ \frac{(C_{t+\tau}^E)^{1-\sigma_C}}{1-\sigma_C} \right\} \quad (7)$$

where C_t^E is the consumption of entrepreneurs.

Firms technology is given by:

$$Y_t = \varepsilon_t^A K_{t-1}^\alpha h_t^{1-\alpha} \quad (8)$$

where ε_t^A is the a technology shock, K_{t-1} is the physical stock of capital with fixed intensity of $\alpha \in [0; 1]$ and h_t is the labor demand.

The balance sheet of the entrepreneur reads as:

$$C_t^E + I_t + w_t h_t + r_{t-1}^E l_{t-1} = Y_t + l_t \quad (9)$$

where I_t is investment, w_t real wage.

The accumulation oh physical capital is given by:

$$I_t \varepsilon_t^I \left(1 - S \left(\frac{I_t}{I_{t-1}} \right) \right) = K_t - (1 - \delta) K_{t-1} \quad (10)$$

where $S(\cdot)$ is an adjustment cost function on investment introduced by [Christiano et al. \(2005\)](#). This function $S(x) = 0.5\kappa(x - \bar{x})^2$ features some congestion cost when entrepreneurs are simultaneously investing at the same time. Parameter $\kappa \geq 0$ is the degree of these adjustment costs. Capital stock is depreciating at a fixed rate of $\delta \in [0, 1]$.

The Lagrangian faced by entrepreneurs reads as:

$$\begin{aligned}\mathcal{L}_t^E = & \sum_{\tau=0}^{\infty} \beta_E^\tau \frac{(C_{t+\tau}^E)^{1-\sigma_C}}{1-\sigma_C} \\ & + \beta_E^\tau \lambda_{t+\tau}^E \left[Y_{t+\tau} + l_{t+\tau} - C_{t+\tau}^E - I_{t+\tau} - w_{t+\tau} h_{t+\tau} - r_{t-1+\tau}^E l_{t-1+\tau} \right] \\ & + \beta_E^\tau \lambda_{t+\tau}^E \varphi_{t+\tau}^E \left[m_K E_t \{q_{t+1+\tau}\} \frac{K_{t+\tau}}{r_{t+\tau}^E} - m_H h_{t+\tau} w_{t+\tau} - l_{t+\tau} \right] \\ & + \beta_E^\tau \lambda_{t+\tau}^E q_{t+\tau} \left[K_{t+\tau} - (1-\delta) K_{t-1+\tau} - I_{t+\tau} \varepsilon_{t+\tau}^I \left(1 - S(I_{t+\tau}/I_{t-1+\tau})^2 \right) \right]\end{aligned}$$

where λ_t^E , φ_t^E and q_t are lagrangian multipliers. Here, we simplify the problem for two last constraints by pre-multiplying to express each constraints into consumption equivalents.

First order conditions are given by:

$$C_t^E : (C_t^E)^{-\sigma_C} - \lambda_t^E = 0 \quad (11)$$

$$l_t : \lambda_t^E - \beta_E E_t \left\{ \lambda_{t+1}^E \right\} r_t^E - \lambda_t^E \varphi_t^E = 0 \quad (12)$$

$$I_t : -\lambda_t^E - \lambda_t^E q_t \varepsilon_t^I \left(1 - \frac{\partial S(I_t/I_{t-1})}{\partial I_t} \right) + \beta_E \lambda_{t+1}^E q_{t+1} \varepsilon_{t+1}^I \frac{\partial S(I_{t+1}/I_t)}{\partial I_t} = 0 \quad (13)$$

$$K_t : \beta_E E_t \left\{ \lambda_{t+1}^E \left[q_{t+1} (1-\delta) + \alpha \frac{Y_{t+1}}{K_t} \right] \right\} - q_t \lambda_t^E + \lambda_t^E \varphi_t^E E_t \{q_{t+1}\} \frac{m_K}{r_t^E} = 0 \quad (14)$$

$$h_t : \left[(1-\alpha) \frac{Y_t}{h_t} - w_t \right] \lambda_t^E - \lambda_t^E \varphi_t^E m_H w_t = 0 \quad (15)$$

Rewriting these conditions dividing by λ_t^E provides:

$$\lambda_t^E = (C_t^E)^{-\sigma_C} \quad (16)$$

$$1 - \varphi_t^E = \beta_E E_t \left\{ \frac{\lambda_{t+1}^E}{\lambda_t^E} \right\} r_t^E \quad (17)$$

$$q_t \varepsilon_t^I = 1 + q_t \varepsilon_t^I \kappa \left(1 + \left(3 \frac{I_t}{I_{t-1}} - 4 \right) \frac{I_t}{I_{t-1}} \right) + \beta_E \left\{ \frac{\lambda_{t+1}^E}{\lambda_t^E} q_{t+1} \varepsilon_{t+1}^I \kappa \left(1 - \frac{I_t}{I_{t-1}} \right) \left(\frac{I_t}{I_{t-1}} \right)^2 \right\} \quad (18)$$

$$r_t^E q_t = (1 - \varphi_t^E) E_t \left\{ \left[q_{t+1} (1-\delta) + \alpha \frac{Y_{t+1}}{K_t} \right] \right\} + \varphi_t^E m_K E_t \{q_{t+1}\} \quad (19)$$

$$w_t = \frac{(1-\alpha) Y_t}{(1 + \varphi_t^E m_H) h_t} \quad (20)$$

The key friction here is the collateral constraint, whose multiplier φ_t^E , provides the shadow cost of one unit of loans and affect in turn the overall discounting of households. When the friction binds a lot, with high value of φ_t^E , households are actually getting more patient and will consume less and invest less, that is mechanically creating the macroeconomic conditions for recession.

1.3 Financial intermediaries

Here, we assume perfectly competitive and frictionless financial intermediaries, which collect deposits to households and provide loans to entrepreneurs:

$$\Pi_t^B = r_t^E l_t + r_t b_t - r_t d_t \quad (21)$$

facing the following balance sheet:

$$d_t = l_t + b_t$$

The first order condition reads as:

$$r_t^E = r_t$$

so there is no wedge between the real lending rate and the real deposit rate.

One of the key contributions of [Gerali et al. \(2010\)](#) and [Iacoviello \(2015\)](#) is to introduce imperfect financial intermediaries.

1.4 Equilibrium conditions

Aggregate consumption reads as:

$$C_t = C_t^E + C_t^H. \quad (22)$$

And total demand is given by:

$$Y_t = C_t + I_t + G_t. \quad (23)$$

1.5 The friction in steady state

The key friction here is the binding constraint. How are we sure that this friction is actually binding? To be sure that the friction is binding, we need to check whether in the deterministic steady state $\bar{\varphi}^E > 0$, if so then the constraint is eternally binding. First, let's remove time subscripts to Euler equations:

$$\begin{aligned} \beta_H \bar{r} - 1 &= 0 \\ 1 - \beta_E \bar{r} - \bar{\varphi}^E &= 0 \end{aligned}$$

Combining these two equations through their common interest rate:

$$\bar{\varphi}^E = 1 - \frac{\beta_E}{\beta_H}$$

The constraint binds as long as $\beta_I/\beta_H < 1$, $\bar{\varphi}^E > 0$.

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